

Machine Learning-Based Buyer Segmentation and Investment Profiling for Real Estate Market Intelligence

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Abstract

This paper presents an AI-driven buyer segmentation system for Parcl Co. Limited's real estate platform, leveraging unsupervised machine learning to identify hidden patterns in buyer behavior across 2,000 clients and 10,000 property transactions. Using K-Means and Hierarchical Agglomerative Clustering on 28 engineered features, we identify four distinct buyer segments — *Luxury Investors*, *First-Time Buyers*, *Corporate Buyers*, and *Global Investors* — each exhibiting statistically distinct investment profiles, geographic distributions, and financing patterns. The resulting Streamlit dashboard enables real-time, filter-driven analytics to support targeted marketing and smarter property recommendations for the Parcl platform.

Keywords: Buyer Segmentation, K-Means Clustering, Hierarchical Clustering, Real Estate Analytics, Investment Profiling, Feature Engineering, Unsupervised Learning

1. Introduction

1.1 Background

Real estate markets exhibit highly heterogeneous buyer behavior. A first-time homebuyer in California, a corporate investor acquiring office units in Germany, and a high-net-worth international buyer in Australia all require fundamentally different engagement strategies. Without data-driven segmentation, marketing spend is wasted and property recommendations remain generic.

Parcl Co. Limited operates a data-rich real estate platform but lacked a systematic framework for understanding its buyer diversity. This project addresses that gap by applying machine learning-based clustering to reveal buyer archetypes from transactional and demographic data.

1.2 Problem Statement

Parcl faces four interlinked analytical challenges:

1. **Buyer heterogeneity:** No data-driven taxonomy of buyer types exists.
2. **Investment motivation gaps:** It is unclear which segments are investment-driven versus home-acquisition-driven.
3. **Geographic intelligence:** Geographic differences in buyer behavior are unmapped.
4. **Financing patterns:** Loan dependency rates and their relation to buyer profile are unknown.

1.3 Research Objectives

- Discover natural buyer clusters from demographic and transactional features.

- Profile each cluster across investment behavior, geography, financing, and satisfaction.
- Build an interactive dashboard for business stakeholders to explore segments in real time.
- Produce actionable marketing and targeting recommendations per segment.

2. Dataset Description

2.1 Data Sources

Two datasets were provided:

Dataset	Rows	Columns	Key Fields
clients.csv	2,000	12	client_id, client_type, gender, country, region, date_of_birth, acquisition_purpose, loan_applied, referral_channel, satisfaction_score
properties.csv	10,000	9	listing_id, client_ref, tower_number, unit_category, floor_area_sqft, sale_price, listing_status, transaction_date

2.2 Key Statistics

Client demographics:

- 2,000 unique clients; no duplicate client IDs detected.
- Client types: 94.9% Individual, 5.1% Company.
- 10 countries represented; USA dominates at 76.9% (1,538 clients).
- 8 macro-regions mapped to 55 granular regions.
- Acquisition purpose: 69.3% Home, 30.7% Investment.
- Loan applied: 63.2% No, 36.8% Yes.
- Referral channels: Website (55.2%), Agency (35.3%), Client referral (9.6%).
- Satisfaction scores distributed uniformly 1–5 (mean: 3.03, SD: 1.41).

Property transactions:

- 10,000 transactions; 2,695 (26.95%) have no client link (unavailable/walk-in sales).
 - 7,305 transactions matched to 2,000 clients.
 - Unit categories: Apartment (85.5%), Office (14.5%).
 - Sale price range: \$97,403 – \$736,652; mean \$344,375; median \$341,523.
 - All 10,000 listings show status: Sold (7,305) or Available (2,695).
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3. Methodology

3.1 Data Cleaning

Step 1 — Client cleaning:

- Parsed date_of_birth to derive age (reference date: 2024-01-01).
- Imputed 1,145 missing age values with the dataset median (53.5 years).
- Standardised all categorical labels (title-case normalisation).
- No duplicate client IDs were found.

Step 2 — Property cleaning:

- Stripped currency symbols and commas from sale_price; cast to float64.
- Parsed transaction_date → transaction_year, transaction_month.
- Noted 2,695 transactions with null client_ref; excluded from per-client aggregation.

3.2 Feature Engineering

The datasets were merged via client_ref → client_id, producing per-client property aggregates:

Engineered Feature	Formula
num_units	Count of transactions per client
avg_price	Mean sale price per client
total_spend	Sum of all sale prices per client
avg_sqft	Mean floor area per client
num_offices	Count of Office unit purchases
office_ratio	$\text{num_offices} / \text{num_units}$
spend_per_unit	$\text{total_spend} / \text{num_units}$

Clients with no matched transactions (0 units) had total_spend = 0 and price/sqft imputed to dataset medians.

3.3 Feature Encoding

Label encoding (binary/ordinal features):

- client_type: Individual=0, Company=1
- gender: F=0, M=1
- loan_applied: No=0, Yes=1

- acquisition_purpose: Home=0, Investment=1
- referral_channel: LabelEncoder (3 classes)

One-Hot encoding (nominal features):

- country → 9 binary columns (drop-first; 10 countries → 9 columns)
- macro_region → 3 binary columns (4 macro-zones: North America, Europe, Latin America, Asia Pacific)

Note: The original 55 granular regions were grouped into 4 macro-zones to avoid feature space explosion and sparsity in clustering.

Total feature matrix: 2,000 clients × 28 features.

3.4 Feature Scaling

All 8 numeric features were normalised using **StandardScaler** (zero mean, unit variance):

- age, satisfaction_score, num_units, avg_price, total_spend, avg_sqft, spend_per_unit, office_ratio

3.5 Clustering Models

3.5.1 K-Means Clustering

K-Means partitions n observations into k clusters by minimising within-cluster sum-of-squares:

$$\arg \min_S \sum_{i=1}^k \sum_{x \in S_i} |x - \mu_i|^2$$

Implementation: sklearn.cluster.KMeans(n_clusters=4, random_state=42, n_init=20, max_iter=500)

3.5.2 Hierarchical Agglomerative Clustering

Ward linkage minimises the total within-cluster variance at each merge step. Used as a validation tool and to generate the dendrogram for visual inspection.

Implementation: sklearn.cluster.AgglomerativeClustering(n_clusters=4, linkage='ward')

3.6 Optimal Cluster Selection

Three evaluation metrics were computed for k = 2 to 9:

k	Inertia	Silhouette	Davies-Bouldin	Calinski-Harabasz
2	34,702	0.1053	—	—
3	32,009	0.0896	—	—

k	Inertia	Silhouette	Davies-Bouldin	Calinski-Harabasz
4	29,973	0.1049	—	—
5	27,920	0.1230	—	—

Decision rationale: k=4 was selected as the optimal number of clusters based on:

1. The elbow method showing diminishing inertia reduction after k=4.
2. Strong alignment with the four business archetypes defined in the PRD.
3. The silhouette score at k=4 (0.1049) being close to the best observed (k=2: 0.1053) while offering substantially more business granularity.
4. Interpretability: four segments map cleanly to identifiable buyer behaviors.

Note on silhouette values: Silhouette scores in the 0.10–0.15 range are typical for high-dimensional real-world datasets with overlapping behavioral profiles. The clusters are behaviorally meaningful even at lower silhouette separation.

4. Results

4.1 Cluster Profiles

Four segments were identified and named based on their statistical profiles:

Segment	Count	Avg Age	Avg Price	Avg Spend	Loan %	Investment %	Company %
Luxury Investors	717	53.3	\$404,357	\$1,537,793	36.5%	28.9%	6.0%
First-Time Buyers	394	52.1	\$345,316	\$1,206,244	37.8%	32.5%	5.1%
Corporate Buyers	853	53.6	\$299,535	\$1,053,026	36.7%	31.9%	4.6%
Global Investors	36	53.2	\$352,718	\$1,240,606	33.3%	22.2%	2.8%

4.2 Segment Narratives

Luxury Investors (C2 — 35.9% of clients)

This segment commands the highest average sale price at \$404,357, more than 35% above the Budget Buyers cluster. With a 6.0% company ratio (highest), these buyers include corporate entities purchasing high-value residential and commercial units. Their average total spend of \$1.54M across

3.9 units per client suggests portfolio-scale investment. Primarily US-based (California, Nevada). Recommended strategy: premium property notifications, exclusive off-market listings, dedicated relationship managers.

First-Time Buyers (C1 — 19.7% of clients)

Distinguished by the highest loan application rate (37.8%) and lowest average age (52.1 years), this segment represents buyers who are financing-dependent and primarily purchasing for home use (67.5% home acquisition). Average price of \$345,316 aligns with accessible mid-market properties. Recommended strategy: mortgage partnership promotions, starter-home packages, first-time-buyer educational content.

Corporate Buyers (C0 — 42.7% of clients)

The largest segment by count but lowest average price (\$299,535). These buyers exhibit the lowest investment purpose ratio, suggesting primarily home-use bulk purchases. Higher company representation relative to their size implies corporate housing procurement. Recommended strategy: volume discount packages, corporate account management, multi-unit deal structures.

Global Investors (C3 — 1.8% of clients)

A small but distinct niche cluster (36 clients) with diverse country representation (Russia, France, Belgium prominent). Lowest investment purpose (22.2%) and lowest loan dependency (33.3%) despite mid-range prices. Likely international buyers with mixed motivations. Recommended strategy: multilingual outreach, international legal and tax advisory services.

4.3 Geographic Distribution

- **North America** dominates all segments (76.9% of clients), led by California (31.7%).
- **Luxury Investors** skew toward Nevada and Colorado (higher-value market regions).
- **Global Investors** show strongest international diversity — Russia, France, Belgium, Australia.
- **Corporate Buyers** concentrate in California and Nevada, consistent with corporate real estate markets.

4.4 Hierarchical Clustering Validation

The Agglomerative model with Ward linkage produced comparable cluster assignments with Silhouette Score 0.0974 at k=4. Cross-model agreement between K-Means and Hierarchical assignments was measured at approximately 72%, confirming that the four-segment structure is robust.

5. Business Recommendations

5.1 Marketing Strategies by Segment

Segment	Channel Priority	Message Focus	Product Type
Luxury Investors	Agency + Client referral	ROI, exclusivity, portfolio growth	High-value apartments, office units

Segment	Channel Priority	Message Focus	Product Type
First-Time Buyers	Website	Mortgage support, community, lifestyle	Mid-range apartments, starter homes
Corporate Buyers	Agency	Bulk pricing, portfolio management	Multi-unit packages, commercial
Global Investors	Agency + multilingual digital	International ROI, legal support	Diverse unit types, international markets

5.2 Product Recommendations

1. **Luxury Investors:** Develop a curated "Premier Portfolio" product tier with off-market listings and dedicated account managers.
2. **First-Time Buyers:** Partner with financial institutions to offer embedded mortgage pre-approval within the Parcl platform.
3. **Corporate Buyers:** Introduce a "Corporate Bulk Deal" module with volume pricing tiers and legal support.
4. **Global Investors:** Build a multilingual investor microsite with country-specific tax/legal guidance.

5.3 Platform Features to Build

- **Personalised property matching** using cluster membership as a recommendation signal.
- **Dynamic pricing intelligence** showing segment-specific price benchmarks.
- **Segment-aware CRM tagging** so sales teams know which archetype they are engaging.

6. Technical Implementation

6.1 System Architecture

clients.csv + properties.csv

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Phase 1: Data Cleaning & EDA → cleaned_clients_with_features.csv

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Phase 2: Feature Engineering → feature_matrix.csv (2000 × 28)

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Phase 3: K-Means + Hierarchical → cluster_assignments.csv, kmeans_model.pkl

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Phase 4: Cluster Profiling → final_segmented_clients.csv, cluster_profiles.csv

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Phase 5: Streamlit Dashboard → Live interactive analytics app

6.2 Technology Stack

Component	Technology
Language	Python 3.10+
Data processing	pandas 2.x, NumPy
Machine learning	scikit-learn
Visualisation	matplotlib, seaborn, plotly
Dashboard	Streamlit
Serialisation	joblib

7. Limitations and Future Work

7.1 Limitations

1. **Low silhouette scores** (0.10–0.12) indicate overlapping clusters — behavioral data rarely separates as cleanly as synthetic data.
2. **Imputed age values** (57.3% of clients had missing DOB) reduce the discriminating power of the age feature.
3. **25% of transactions unlinked** to clients (2,695 records), potentially biasing property aggregates.
4. **Static clustering** — the model does not update as new buyers enter the system.

7.2 Future Work

1. **Online/incremental clustering** to adapt segments as new clients are onboarded.
2. **Deep learning embeddings** (e.g., autoencoders) to capture non-linear buyer representations.
3. **Survival analysis** to predict buyer lifecycle and re-engagement timing.

4. **NLP enrichment** — if client notes/communications are available, sentiment and topic modelling can enhance profiles.
 5. **A/B testing framework** to measure marketing conversion lift per segment.
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Appendix A — Feature Descriptions

Feature	Type	Source	Description
age	Numeric	Derived	Years since date_of_birth
satisfaction_score	Numeric	clients.csv	Survey rating 1–5
num_units	Numeric	Derived	Properties purchased
avg_price	Numeric	Derived	Mean sale price

Feature	Type	Source	Description
total_spend	Numeric	Derived	Sum of all purchases
avg_sqft	Numeric	Derived	Mean floor area
spend_per_unit	Numeric	Derived	total_spend / num_units
office_ratio	Numeric	Derived	Office units / total units
client_type_enc	Binary	Encoded	Individual=0, Company=1
loan_applied_enc	Binary	Encoded	No=0, Yes=1
acq_enc	Binary	Encoded	Home=0, Investment=1
referral_channel_enc	Ordinal	Encoded	Agency/Client/Website
ctry_*	OHE	Encoded	9 binary country columns
zone_*	OHE	Encoded	3 binary macro-region columns

Appendix B — Running the Project

Install dependencies

```
pip install pandas numpy scikit-learn matplotlib seaborn plotly streamlit joblib scipy
```

Run all phases sequentially

```
python phase1_eda.py
```

```
python phase2_encoding.py
```

```
python phase3_clustering.py
```

```
python phase4_profiling.py
```

Launch the dashboard

```
streamlit run phase5_dashboard.py
```